INCH-POUND
A-A-60062A
May 20, 2025
SUPERSEDING
A-A-60062
May 9, 2024

COMMERCIAL ITEM DESCRIPTION

STUDS, CADMIUM AND HEXAVALENT CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers studs containing no cadmium or hexavalent-chromium in their finishes.

1.1 Intended use. Studs contained herein are intended for ground vehicle applications. Studs described in this CID shall not be used for aerospace applications without design authority approval.

1.2 Clean finish. See Section 7 (paragraph 7.3) for definition of clean finish when referenced on a table.

2 CLASSIFICATION

2.1 Types.

Type I	-	AIA/NAS184 - Stud, Fine Thread
Type II	-	AIA/NASM51864 - Stud, Plain, General Purpose
Type III	-	AIA/NASM51833 – Stud, Locked In, Key-Locked, Lightweight

3 SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the studs shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed above by type with the exception of, the plating/finishes shall be free of cadmium and hexavalent chromium finishes as specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

3.2 Dimensions. Dimensions for all designated types of studs apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent chromium-free Part or Identifying Number (PIN) codes and finish/plating options. PIN finish/plating codes for all designated types of studs:

QX = Carbon/alloy steel shall be zinc-nickel plated in accordance MIL-PRF-32647, Type K16.
P5 = CRES shall be cleaned, descaled and passivated in accordance with SAE AMS2700, Method 1, Type 6, 7, or 8, or Method 2. Hexavalent chromium containing compounds shall be prohibited.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of the address information using the ASSIST Online database at https://assist.dla.mil.

FSC 5307

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3.4 Lubrication. Lubrication types (if any) for each type of stud shall be specified in each AIA/NASM specification (see 2.1).

3.5 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent chromium with the cadmium and hexavalent chromium-free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering (PIN) procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. The following finish/plating PIN's are cadmium and hexavalent chromium-free for designated stud types.

TABLE I. Type I (-1) stud, fine thread.

(Use NAS184 for material, sizes/lengths, and heat treatment 125 to 145 KSI per SAE AMS2759.)

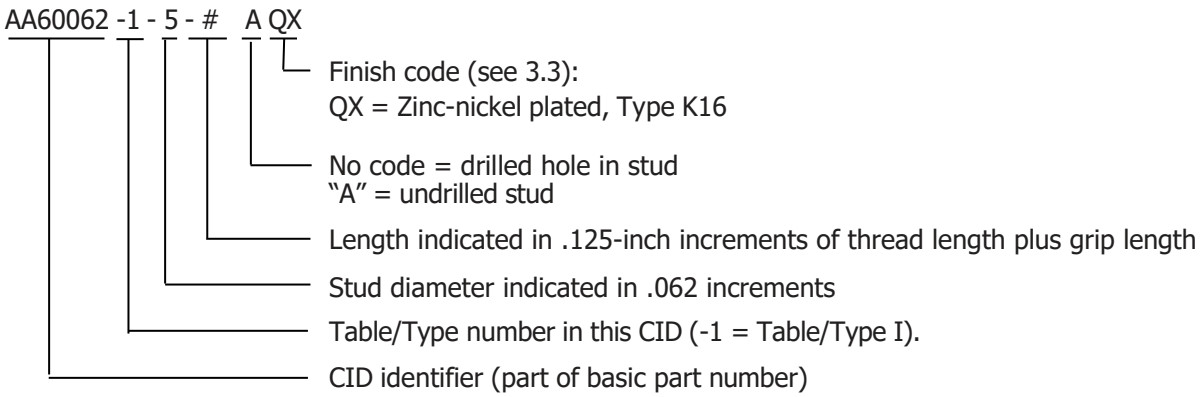
A-A-60062 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NAS184 Referenced "E Thread" Cadmium Plated, DO NOT USE on Ground Vehicles	NAS184 Stud Diameter Dash Number
AA60062-1-3-#*	.1900-32	3
AA60062-1-4-#*	.2500-28	4
AA60062-1-5-#*	.3125-24	5
AA60062-1-6-#*	.3750-24	6
AA60062-1-7-#*	.4375-20	7
AA60062-1-8-#*	.5000-20	8

NOTES:

Part number for clean finished studs in this CID shall be as described in the "PIN construction" (see next page).

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PIN construction:



Examples of clean finish PIN's:

AA60062-1-5-23QX = a NAS184 – stud, Ø.312, length 2.875, drilled hole in stud, zinc-nickel plated, Type K16.

AA60062-1-5-23AQX = a NAS184 – stud, Ø.312, length 2.875, undrilled stud, zinc-nickel plated, Type K16.

TABLE II. Type II (-2) stud, plain, general purpose.

(Use NASM51864 for material and sizes/lengths.)

A-A-60062 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM51864 Reference Part Number Cadmium Plated, DO NOT USE on Ground Vehicles	A-A-60062 Part Number Passivated (CRES) (Clean Finish)	NASM51864 Reference Part Number Passivated (CRES), DO NOT USE on Ground Vehicles
AA60062-2-102-#*	MS51864-102-#	AA60062-2-202-#*	MS51864-202-#
AA60062-2-103-#*	MS51864-103-#	AA60062-2-203-#*	MS51864-203-#
AA60062-2-104-#*	MS51864-104-#	AA60062-2-204-#*	MS51864-204-#
AA60062-2-105-#*	MS51864-105-#	AA60062-2-205-#*	MS51864-205-#
AA60062-2-106-#*	MS51864-106-#	AA60062-2-206-#*	MS51864-206-#
AA60062-2-107-#*	MS51864-107-#	AA60062-2-207-#*	MS51864-207-#
AA60062-2-108-#*	MS51864-108-#	AA60062-2-208-#*	MS51864-208-#
AA60062-2-109-#*	MS51864-109-#	AA60062-2-209-#*	MS51864-209-#
AA60062-2-110-#*	MS51864-110-#	AA60062-2-210-#*	MS51864-210-#
AA60062-2-111-#*	MS51864-111-#	AA60062-2-211-#*	MS51864-211-#
AA60062-2-112-#*	MS51864-112-#	AA60062-2-212-#*	MS51864-212-#
AA60062-2-113-#*	MS51864-113-#	AA60062-2-213-#*	MS51864-213-#

NOTES:

(Brass and nickel-copper alloy per NASM51864 do not apply to this CID.)

Material, carbon/alloy steel per SAE AIR4127; CRES per SAE AIR4127, 300 series; mechanical properties per FF-S-1362.

Part number for clean finished studs in this CID shall be as described in the "PIN construction" (see next page).

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PIN construction:

AA60062 -2 -102-# *

- Finish Code (see 3.3):
  QX = Zinc-nickel plated, Type K16
  P5 = Passivated
- Stud Length dash number (see Table III below)
- Stud Diameter dash number (see Table III below)
- Table/Type number in this CID (-2 = Table/Type II).
- CID identifier (part of basic part number)

Table III. Length and diameter dash numbers.

Length Dash No.	Diameter Dash Number from NASM51864											
	102	103	104	105	106	107	108	109	110	111	112	113
	202	203	204	205	206	207	208	209	210	211	212	213
	Overall Stud length											
12	1.500	--	--	--	--	--	--	--	--	--	--	--
14	1.750	1.750	--	--	--	--	--	--	--	--	--	--
16	2.000	2.000	2.000	--	--	--	--	--	--	--	--	--
18	2.250	2.250	2.250	2.250	--	--	--	--	--	--	--	--
20	2.500	2.500	2.500	2.500	2.500	--	--	--	--	--	--	--
22	2.750	2.750	2.750	2.750	2.750	2.750	--	--	--	--	--	--
24	3.000	3.000	3.000	3.000	3.000	3.000	3.000	--	--	--	--	--
26	--	--	3.250	3.250	3.250	3.250	3.250	--	--	--	--	--
28	--	--	3.500	3.500	3.500	3.500	3.500	3.500	--	--	--	--
30	--	--	3.750	3.750	3.750	3.750	3.750	3.750	--	--	--	--
32	--	--	4.000	4.000	4.000	4.000	4.000	4.000	4.000	--	--	--
34	--	--	--	--	4.250	4.250	4.250	4.250	4.250	--	--	--
36	--	--	--	--	4.500	4.500	4.500	4.500	4.500	4.500	--	--
38	--	--	--	--	4.750	4.750	4.750	4.750	4.750	4.750	4.750	--
40	--	--	--	--	5.000	5.000	5.000	5.000	5.000	5.000	5.000	--
44	--	--	--	--	--	--	5.500	5.500	5.500	5.500	5.500	5.500
48	--	--	--	--	--	--	6.000	6.000	6.000	6.000	6.000	6.000
52	--	--	--	--	--	--	--	--	--	--	6.500	6.500
56	--	--	--	--	--	--	--	--	--	--	7.000	7.000
60	--	--	--	--	--	--	--	--	--	--	7.500	7.500
64	--	--	--	--	--	--	--	--	--	--	8.000	8.000

Examples of clean finish PIN's:

AA60062-2-102-16QX = a NASM51864 – stud, carbon or alloy steel, Ø.2500 inches, 2.000 inches overall stud length, zinc-nickel Type K16 plated.
AA60062-2-204-16P5 = a NASM51864 – stud, CRES, Ø.3750 inches, 2.000 inches overall stud length, passivated.

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TABLE IV. Type III (-3) stud, locked in, key-locked, lightweight.

(Use NASM51833 for material, heat treatment (excluding 303 and 303SE CRES), and sizes/lengths.)

Coarse Thread, Nut End				
A-A-60062 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60062 Part Number Passivated (CRES) (Clean Finish)	NASM51833 Reference Part Number DO NOT USE on Ground Vehicles	Nut End Thread (see NASM51833)	Stud End Thread (see NASM51833)
AA60062-3-101#^QX	AA60062-3*101#^P5	MS51833-101#^	.1900-24	.3125-18 UNC-2A
AA60062-3-102#^QX	AA60062-3*102#^P5	MS51833-102#^	.2500-20	.3750-16 UNC-2A
AA60062-3-103#^QX	AA60062-3*103#^P5	MS51833-103#^	.3125-18	.4375-14 UNC-2A
AA60062-3-104#^QX	AA60062-3*104#^P5	MS51833-104#^	.3750-16	.5000-13 UNC-2A
AA60062-3-105#^QX	AA60062-3*105#^P5	MS51833-105#^	.4375-14	.5625-12 UNC-2A
AA60062-3-106#^QX	AA60062-3*106#^P5	MS51833-106#^	.5000-13	.6250-11 UNC-2A

Fine Thread, Nut End				
A-A-60062 Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	A-A-60062 Part Number Passivated (CRES) (Clean Finish)	NASM51833 Reference Part Number DO NOT USE on Ground Vehicles	Nut End Thread (see NASM51833)	Stud End Thread (see NASM51833)
AA60062-3-201#^QX	AA60062-3*201#^P5	MS51833*201#^	.1900-32	.3125-18 UNC-2A
AA60062-3-202#^QX	AA60062-3*202#^P5	MS51833*202#^	.2500-28	.3750-16 UNC-2A
AA60062-3-203#^QX	AA60062-3*203#^P5	MS51833*203#^	.3125-24	.4375-14 UNC-2A
AA60062-3-204#^QX	AA60062-3*204#^P5	MS51833*204#^	.3750-24	.5000-13 UNC-2A
AA60062-3-205#^QX	AA60062-3*205#^P5	MS51833*205#^	.4375-20	.5625-12 UNC-2A
AA60062-3-206#^QX	AA60062-3*206#^P5	MS51833*206#^	.5000-20	.6250-11 UNC-2A

NOTES: * for material since two options are possible for CRES, # for drilled hole or not since two options are available, and ^ for length dash number since numerous options exist (see Table V also)

Part number for clean finished studs in this CID shall be as described in the "PIN construction" below

AA60062 -3 * 102 D 24 QX

- QX: Finish Code (see 3.3):
  QX = Zinc-nickel plated, Type K16
  P5 = Passivated
- 24: Length dash number (see Table V below)
- D: Add "D" instead of "-" for drilled hole
- 102: Dash number (see Table IV above)
- *: Material (refer to NASM51833)
  "-" = Alloy steel, grade 4140 or 8740
  "A" = A286 CRES
  "C" = 303 or 303SE CRES
- -3: Table/Type number in this CID (-3 = Table/Type III).
- AA60062: CID identifier (part of basic part number)

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Table V. Tabulated lengths.

L Nut End ±.030	Length Dash Number					
	Nominal Diameter					
	.190	.250	.3125	.375	.4375	.500
.375	6	6				
.438	7	7				
.500	8	8				
.562	9	9	9			
.625	10	10	10	10	10	
.688	11	11	11	11	11	
.750	12	12	12	12	12	12
.812	13	13	13	13	13	13
.875	14	14	14	14	14	14
.938	15	15	15	15	15	15
1.000	16	16	16	16	16	16
1.062	17	17	17	17	17	17
1.125	18	18	18	18	18	18
1.188	19	19	19	19	19	19
1.250	20	20	20	20	20	20
1.312	21	21	21	21	21	21
1.375	22	22	22	22	22	22
1.438	23	23	23	23	23	23
1.500	24	24	24	24	24	24
1.562	25	25	25	25	25	25
1.625	26	26	26	26	26	26
1.688	27	27	27	27	27	27
1.750	28	28	28	28	28	28
1.812	29	29	29	29	29	29
1.875	30	30	30	30	30	30
1.938	31	31	31	31	31	31
2.000	32	32	32	32	32	32
2.125	34	34	34	34	34	34
2.250	36	36	36	36	36	36
2.375	38	38	38	38	38	38
2.500	40	40	40	40	40	40
2.625		42	42	42	42	42
2.750		44	44	44	44	44
2.875		46	46	46	46	46
3.000		48	48	48	48	48
3.125		50	50	50		50
3.250		52	52	52		52
3.375		54	54	54		54
3.500		56	56	56		56
3.625		58	58	58		58
3.750		60	60	60		60
3.875		62	62	62		62
4.000		64	64	64		64

Note: See note 12 in NASM51833 for thread length of parts above thicker black line.

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Examples of clean finish PIN's:

AA60062-3-105-16QX = a NASM51833 – stud, alloy steel grade 4140 or 8740, .4375-14 nut end thread, .5625-12 UNC-2A stud end thread, 1.000 length of nut end, zinc-nickel Type K16 plated.
AA60062-3-105D16QX = a NASM51833 – stud, alloy steel grade 4140 or 8740, .4375-14 nut end thread, .5625-12 UNC-2A stud end thread, 1.000 length of nut end with drilled hole, zinc-nickel Type K16 plated.
AA60062-3A205-28P5 = a NASM51833 – stud, A286 CRES, .4375-20 nut end thread, .5625-12 UNC-2A stud end thread, 1.750 length of nut end, passivated.
AA60062-3C205D28P5 = a NASM51833 – stud, 303 or 303SE CRES, .4375-20 nut end thread, .5625-12 UNC-2A stud end thread, 1.750 length of nut end with drilled hole, passivated.

7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/.

7.2.2 Executive order (EO). Copies of EO 14057 are available online from https://www.fedcenter.gov/programs/eo14057/.

7.2.3 Military specifications. Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 AIA documents. Copies of Aerospace Industries Association documents are available online from https://www.aia-aerospace.org/.

7.2.5 SAE documents. Copies of SAE International documents are available online from https://www.sae.org/.

7.3 Clean finish definition. In the tables, where it states "clean finish," clean finish is defined as treatments which do not include more than 0.1 percent by weight Cadmium or Hexavalent Chromium in any homogenous layer.

7.4 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. National stock number (if available).
c. Material.
d. Hexavalent chromium or cadmium free part number (PIN) from this CID.
e. Quantity.

7.5 Key words:

Alloy steel
Carbon steel
Clean finish
CRES
Plate
Plating
Steel
Zinc-nickel

7.6 Changes from previous issue. The margins of this CID are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – AT
Navy – SH
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – CG, YD
Air Force – 03, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

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